COOLSHADE · REGULATORY AND STANDARDS REGISTER

CoolShade regulatory and standards register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coolshade		
Source	docs/07-regulatory.md @ ae091ec		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-30	Amish Chadha	First issue in the kit 1.9.1 upgrade, for the current design (option A) and the first site, a market lane in a hot, dry city; every applicability is to assess with a qualified reviewer

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No product certification is sought at TRL 3. The rules that matter most for a public misting shade are local and about people, not products: water hygiene for a mist that people breathe (G1), permission to stand a structure on public land (G2), wind loading of a canopy (G3) and the anchors in the slab (G6). The first site is a market lane in a hot, dry city, with the Phoenix area as the first candidate (CSH-DEC-001); nothing has been agreed with any authority, so the jurisdiction column names the likely rule and every applicability is "to assess". Not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	Local public health rules on water systems that make aerosols, and the WHO guidance on *Legionella*	City or county of the site; international guidance	Control of *Legionella* in water at 20 to 50 °C; temperature, water age, cleaning, sampling	The 7 bar mist is an aerosol from 34 °C tank water (CSH-CAL-001 I3); PR-009 is at risk	Market operator and health authority	To assess; a hygiene plan is a pre-condition of a public trial (CSH-DEC-001 to-confirm item 11)
G2	Permission to install a structure on public land or in a market	City or market operator	Right to stand a fixed canopy; setback from vehicles and walkways; sight lines	Every site; the street-side posts stand 25 mm inside the roof's street edge (CSH-DDR-003 P1)	Market operator	To assess
G3	Wind load rules for canopies and open structures (for example EN 1991-1-4 in Europe, ASCE 7 in the United States)	Country or state of the site	Design gust speed, pressure coefficients, importance factors	CSH-CAL-001 section G uses 30 m/s, a net pressure coefficient of 1.2 and a 1.5 factor; the site's design gust may differ	Design lead	To assess (number and title only; text not consulted)
G4	EN 1090, execution of steel structures	European Union; reference elsewhere	Fabrication and assembly quality of steel structures	The bolted 80 x 80 x 3 mm frame; welding appears only in options B and C	Design lead	To assess
G5	ISO 4017, ISO 4032 and ISO 7089, hexagon bolts, nuts and plain washers	International	Dimensions of the fasteners	Every bolt, nut and washer in the D3 model uses these dimensions (`.kit/cad_parts.py`)	Design lead	Met by design in the model
G6	European Assessment Document EAD 330232, mechanical fasteners for use in concrete	European Union; reference elsewhere	Anchor performance in cracked and uncracked concrete	The four M16 anchor sets at the post bases (about 2.8 kN each with the cloth off, CSH-CAL-001 G13)	Site engineer	To assess (the slab and anchor type are unknown until a site is chosen)
G7	IEC 60529, degrees of protection provided by enclosures (IP code)	International	Ingress of dust and water into the controller enclosure	The enclosure is specified IP65 (PR-012)	Design lead	To assess (the maker's rating to confirm)
G8	IEC 62619, safety requirements for secondary lithium cells and batteries	International	Safety of the LiFePO4 battery	One 12.8 V 25 Ah battery with a BMS and a cold-charge cutoff	Design lead	To assess (maker's declaration to confirm)

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	for industrial applications					
G9	United Nations Manual of Tests and Criteria, part III, section 38.3	International (transport)	Transport tests for lithium batteries	Shipping the battery to a builder	Builder	To assess
G10	IEC 61215 and IEC 61730, photovoltaic module qualification and safety	International	The 100 W solar panel	A bought panel; confirm the maker's certificates	Builder	To assess
G11	Local electrical rules for extra-low-voltage DC installations	Country or state of the site	Fusing, cable sizes, isolation of a 12 V system	Battery fused at the terminal (15 A, margin 2.0, CSH-CAL-001 I2)	Builder	To assess
G12	Backflow prevention rules for a mains water connection	City water utility	A mains connection to the tank, if used	The mains connection is optional (CSH-DDR-001 D4) and needs a backflow preventer	Market operator	To assess
G13	Accessible design rules for clear space and routes (the common clear floor space is 30 by 48 in, or about 0.76 x 1.22 m)	Country of the site	A wheelchair space and a route to it	PR-001 requires a 0.8 x 1.2 m space; the posts stand on a 2,800 x 2,350 mm grid	Design lead	To assess
G14	ISO 11228-1, manual handling: lifting and carrying	International	Mass limits for manual lifting	The heaviest piece is a front post, 21.0 kg; no part over 40 kg (PR-017)	Design lead	Met by design (two-person lift of each piece)
G15	Rules on public misting, pressure equipment and water use in a drought	City, county or state of the site	Whether a 7 bar pump system and its water use need a permit	A 12 V, 60 W diaphragm pump at about 7 bar; 85 L per design day	Market operator	To assess
G16	ISO 12100, general principles for risk assessment	International	Method of the hazard analysis	CSH-RSK-001 follows it	Design lead	Applies as a method

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- World Health Organization. "Legionellosis." https://www.who.int/news-room/fact-sheets/detail/legionellosis (checked 2026-10-30), as in CSH-PRB-001.
- World Health Organization. "Climate change and health: heat." https://www.who.int/news-room/fact-sheets/detail/climate-change-heat-and-health (checked 2026-10-30), as in CSH-PRB-001.
- Standards G3 to G5, G7 to G11, G13, G14 and G16 are cited by number and title or description only; their texts were not consulted for this register and each applicability is to be confirmed by the role named. No standard number is given where none was known.